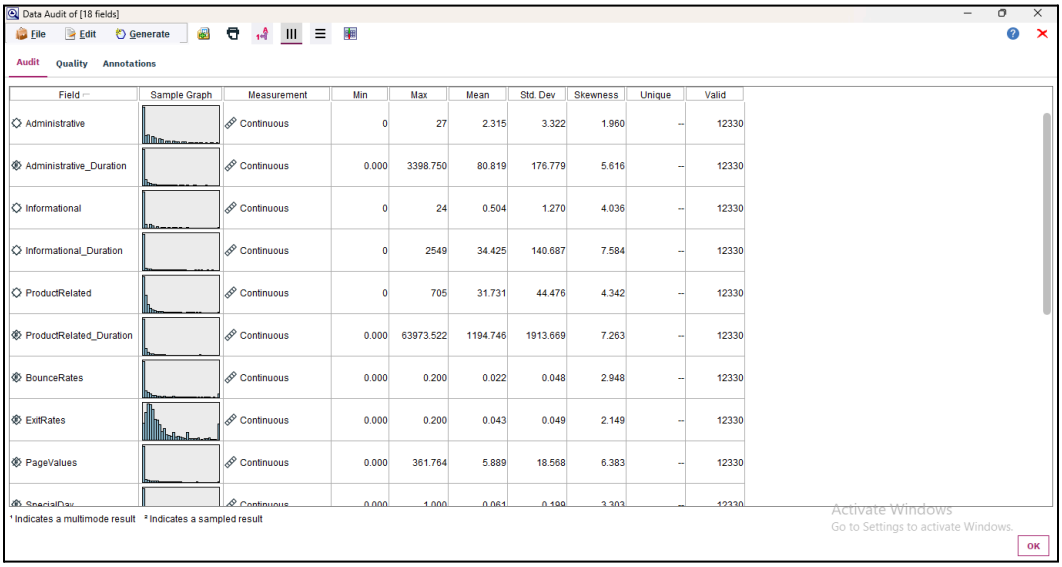
Predictive E-Commerce Cart Abandonment Project

1. Project Overview

This project focuses on predicting e-commerce cart abandonment and identifying the key drivers of successful checkouts. By analyzing customer browsing behavior, operating systems, and page values, the project leverages machine learning to help businesses understand why users leave without purchasing.

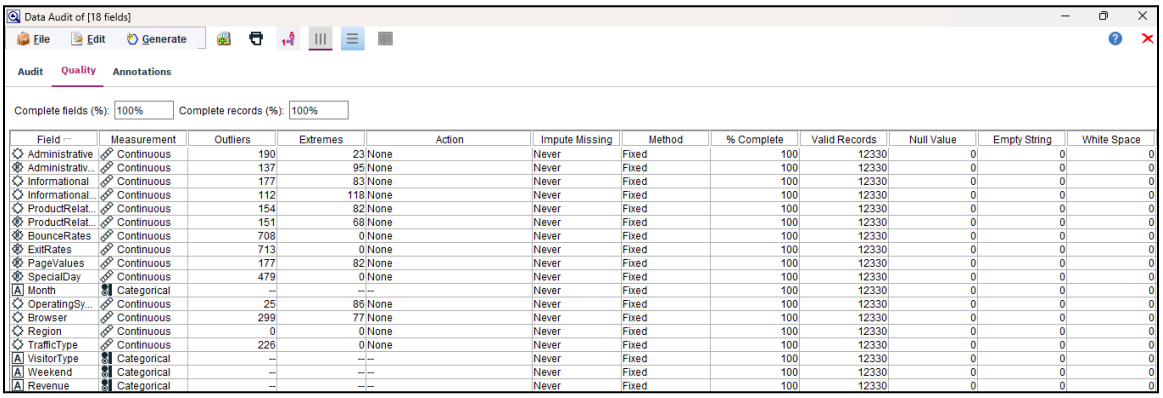
2. Data Preparation (IBM SPSS Modeler)

IBM SPSS Modeler was utilized as the core data science engine to prepare the dataset following the CRISP-DM methodology. SPSS allowed for visual, node-based data engineering and feature creation.



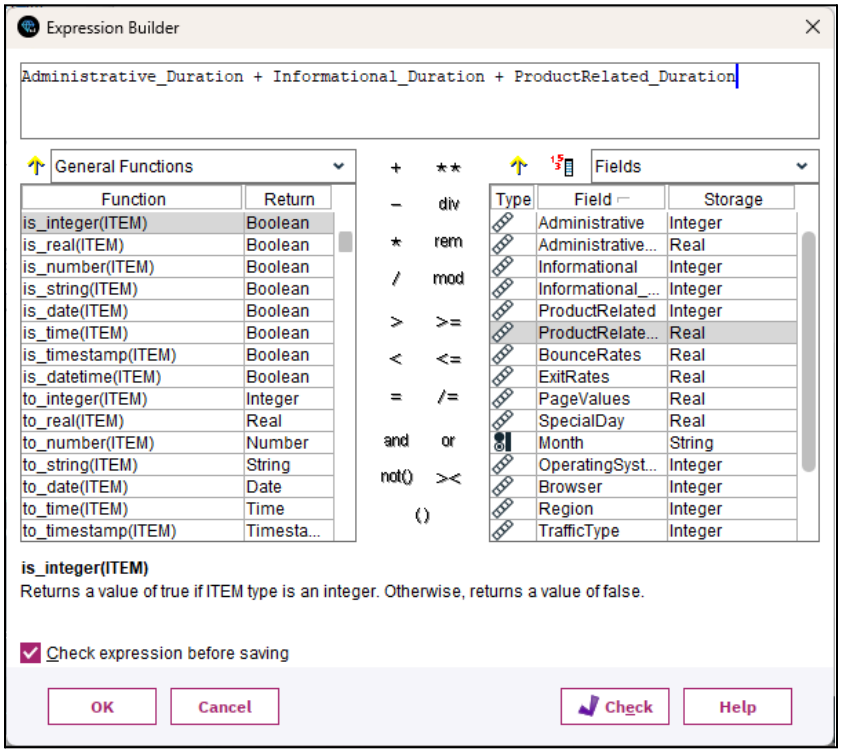
Field	Sample Graph	Measurement	Min	Max	Mean	Std. Dev	Skewness	Unique	Valid
Administrative		Continuous	0	27	2.315	3.322	1.960	--	12330
Administrative_Duration		Continuous	0.000	3398.750	80.819	176.779	5.616	--	12330
Informational		Continuous	0	24	0.504	1.270	4.036	--	12330
Informational_Duration		Continuous	0	2549	34.425	140.687	7.584	--	12330
ProductRelated		Continuous	0	705	31.731	44.476	4.342	--	12330
ProductRelated_Duration		Continuous	0.000	63973.522	1194.746	1913.669	7.263	--	12330
BounceRates		Continuous	0.000	0.200	0.022	0.048	2.948	--	12330
ExitRates		Continuous	0.000	0.200	0.043	0.049	2.149	--	12330
PageValues		Continuous	0.000	361.764	5.889	18.568	6.383	--	12330
SpecialDay		Continuous	0.000	1.000	0.061	0.199	3.393	--	12330

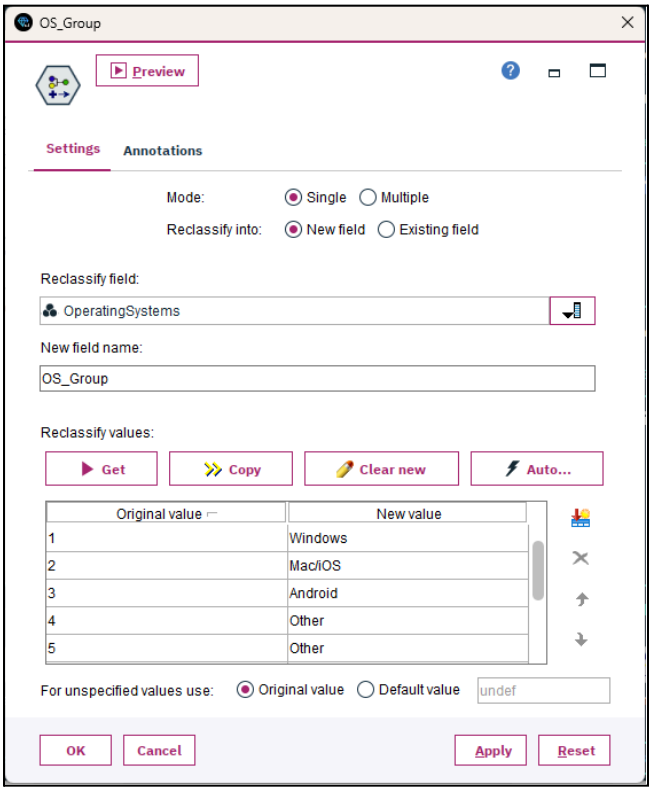
* Indicates a multimode result ² Indicates a sampled result

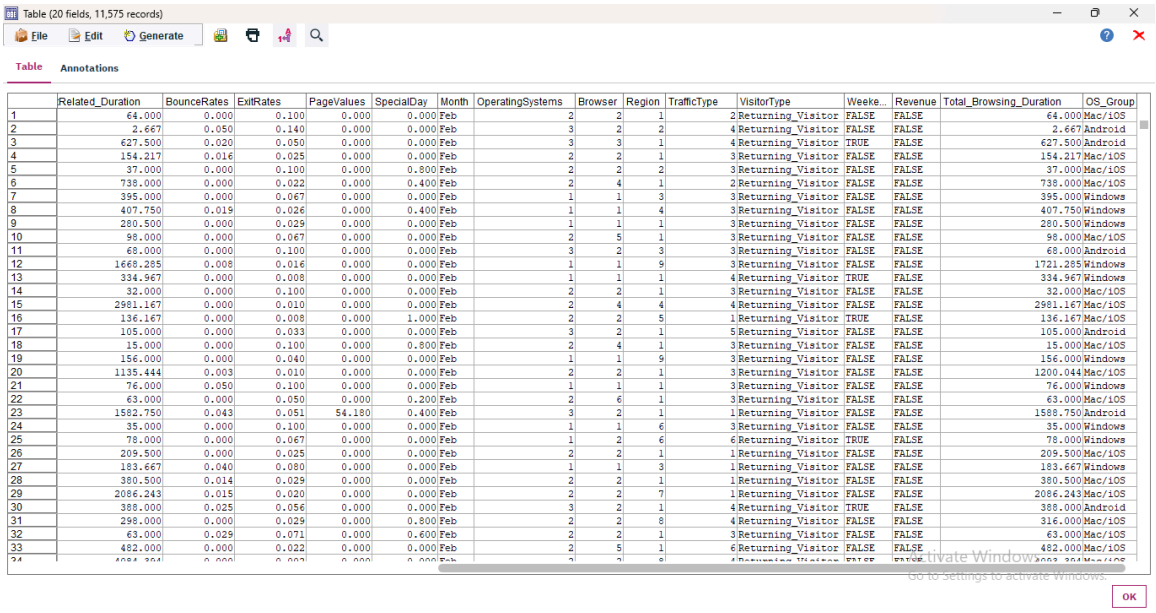


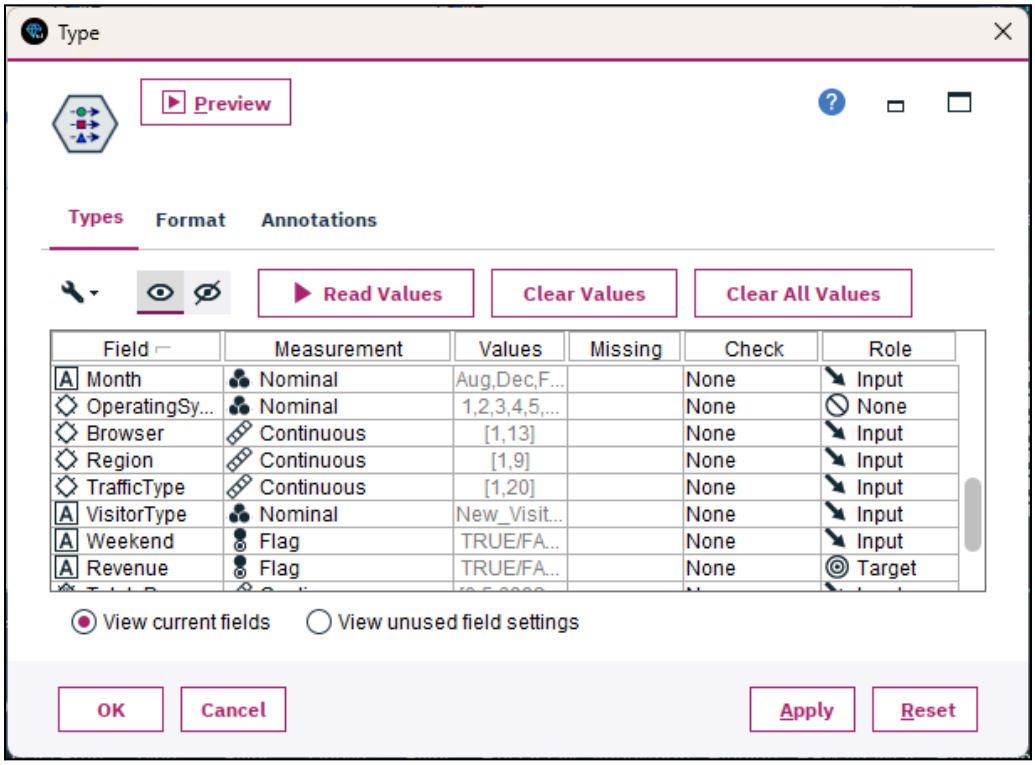
Field	Measurement	Outliers	Extremes	Action	Impute Missing	Method	% Complete	Valid Records	Null Value	Empty String	White Space
Administrative	Continuous	190	23	None	Never	Fixed	100	12330	0	0	0
Administrativ...	Continuous	137	95	None	Never	Fixed	100	12330	0	0	0
Informational	Continuous	177	83	None	Never	Fixed	100	12330	0	0	0
Informational...	Continuous	112	118	None	Never	Fixed	100	12330	0	0	0
ProductRelat...	Continuous	154	82	None	Never	Fixed	100	12330	0	0	0
ProductRelat...	Continuous	151	68	None	Never	Fixed	100	12330	0	0	0
BounceRates	Continuous	708	0	None	Never	Fixed	100	12330	0	0	0
ExitRates	Continuous	713	0	None	Never	Fixed	100	12330	0	0	0
PageValues	Continuous	177	82	None	Never	Fixed	100	12330	0	0	0
SpecialDay	Continuous	479	0	None	Never	Fixed	100	12330	0	0	0
Month	Categorical	--	--	--	Never	Fixed	100	12330	0	0	0
OperatingSy...	Continuous	25	86	None	Never	Fixed	100	12330	0	0	0
Browser	Continuous	299	77	None	Never	Fixed	100	12330	0	0	0
Region	Continuous	0	0	None	Never	Fixed	100	12330	0	0	0
TrafficType	Continuous	226	0	None	Never	Fixed	100	12330	0	0	0
VisitorType	Categorical	--	--	--	Never	Fixed	100	12330	0	0	0
Weekend	Categorical	--	--	--	Never	Fixed	100	12330	0	0	0
Revenue	Categorical	--	--	--	Never	Fixed	100	12330	0	0	0

Key Feature Engineering: Created a custom 'Total_Browsing_Duration' field aggregating administrative, informational, and product-related durations, and grouped Operating Systems into meaningful categories (Mac/iOS, Windows, Android, Other).



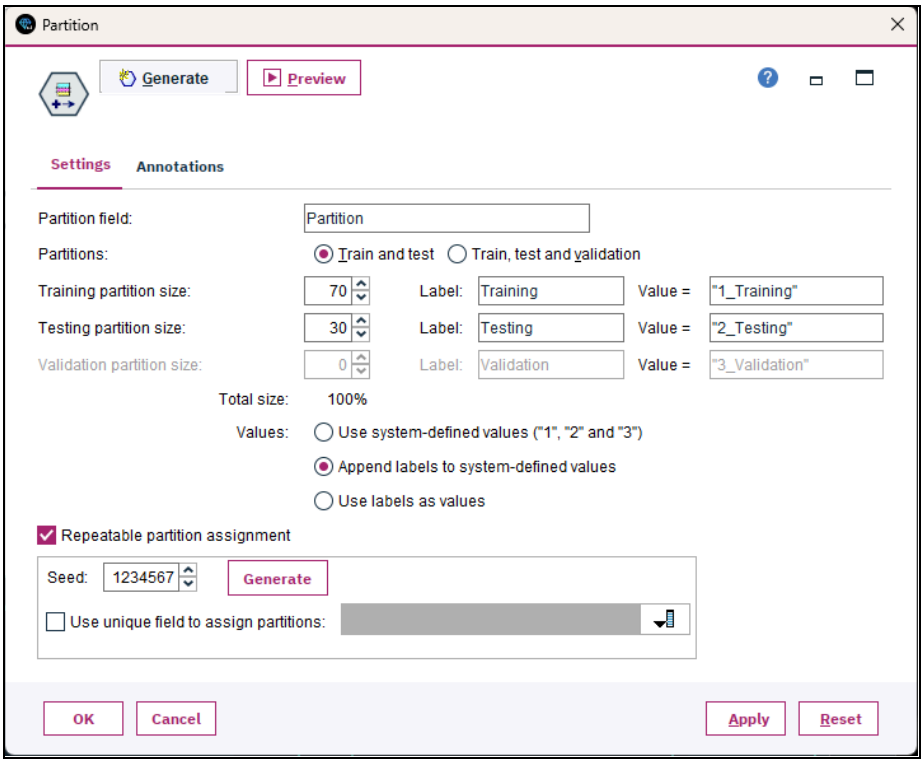




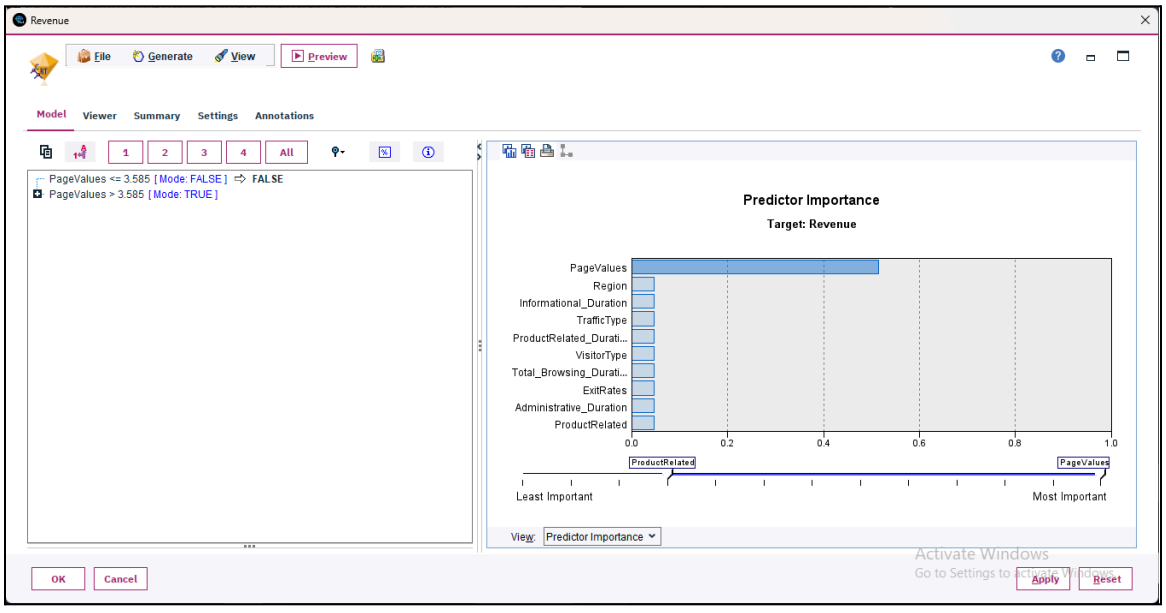


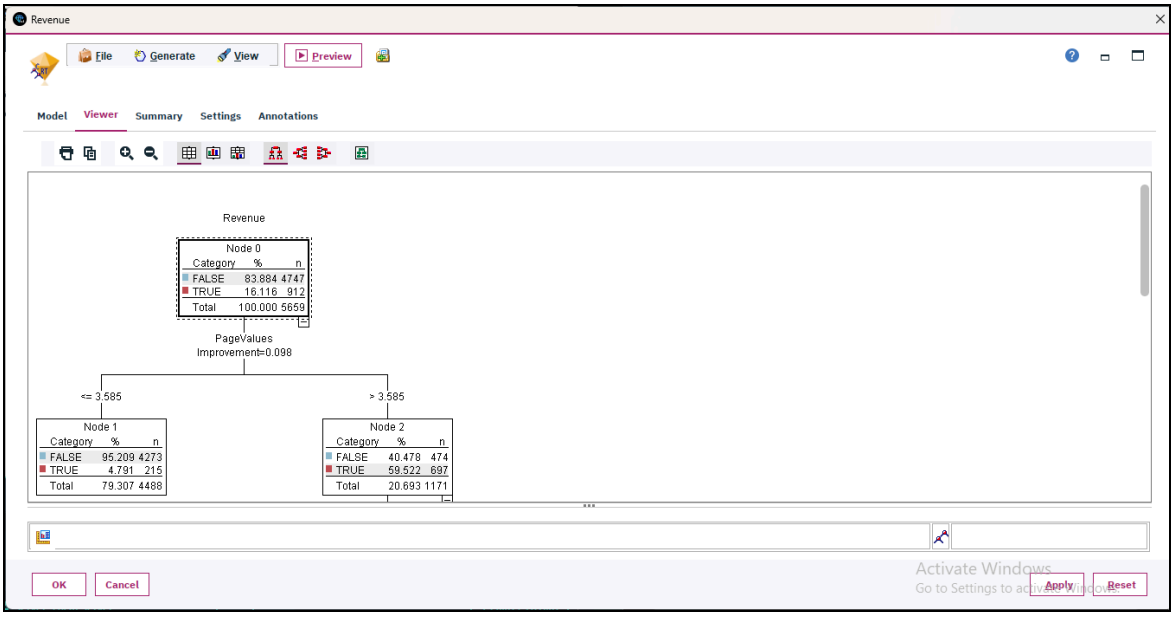
3. Predictive Modeling & Evaluation

The data was partitioned into 70% Training and 30% Testing sets. Three primary models were trained and compared to predict checkout success (Revenue): Classification and Regression (C&R) Tree, Logistic Regression, and Neural Network.



Key Insight: The evaluation metrics and C&R Tree revealed that 'PageValues' is the single most important predictor of whether a user will complete a purchase.





Analysis of [Revenue]

File Edit

Analysis Annotations

Collapse All Expand All

Results for output field Revenue

Individual Models

Comparing $R-Revenue with Revenue

'Partition'	1_Training		2_Testing	
Correct	7,286	90.15%	3,118	89.26%
Wrong	796	9.85%	375	10.74%
Total	8,082		3,493	

Comparing $L-Revenue with Revenue

'Partition'	1_Training		2_Testing	
Correct	7,096	87.8%	3,062	87.66%
Wrong	986	12.2%	431	12.34%
Total	8,082		3,493	

Comparing $N-Revenue with Revenue

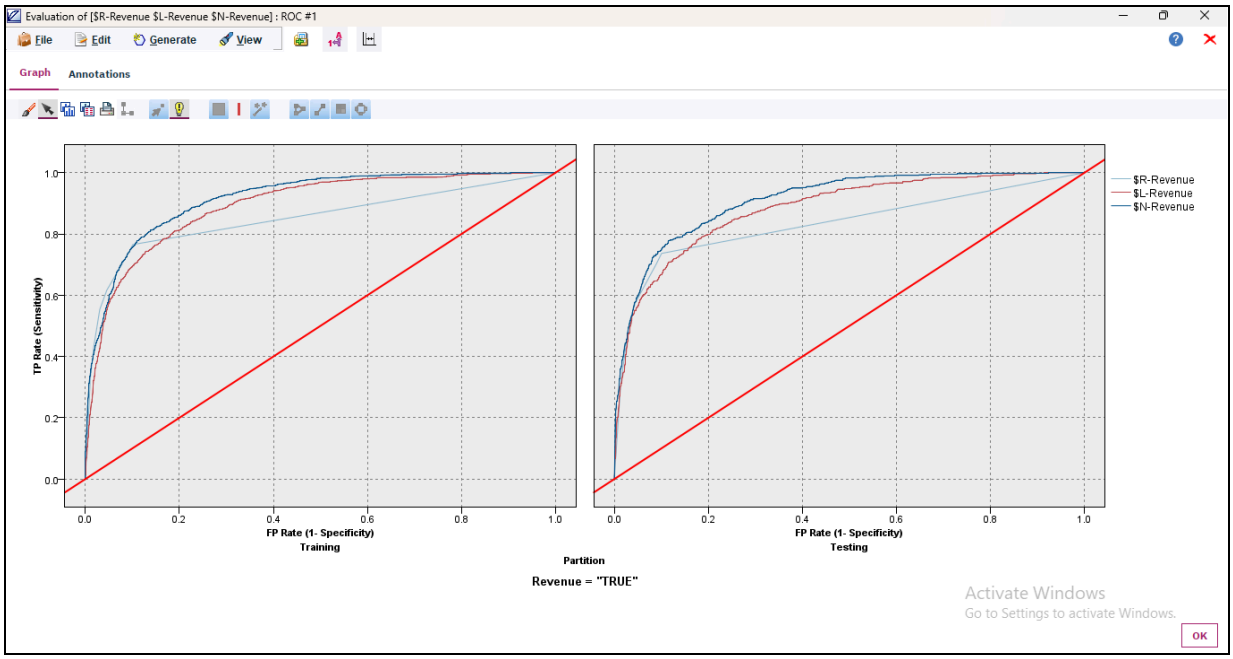
'Partition'	1_Training		2_Testing	
Correct	7,196	89.04%	3,116	89.21%
Wrong	886	10.96%	377	10.79%
Total	8,082		3,493	

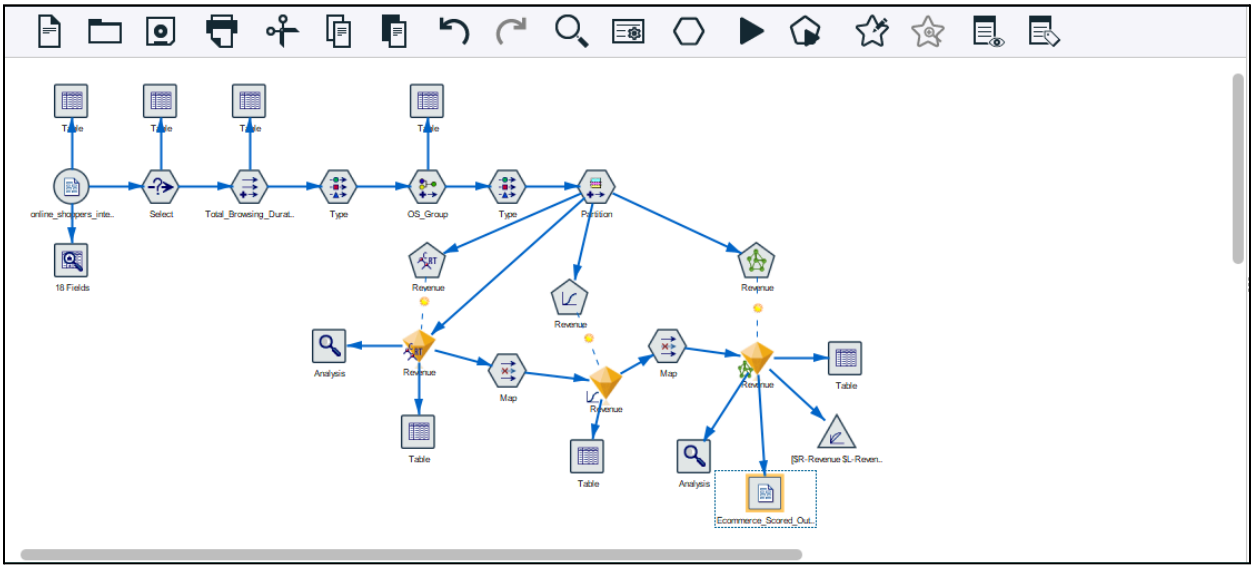
Agreement between $R-Revenue $L-Revenue $N-Revenue

'Partition'	1_Training		2_Testing	
Agree	7,322	90.6%	3,197	91.53%
Disagree	760	9.4%	296	8.47%
Total	8,082		3,493	

Comparing Agreement with Revenue

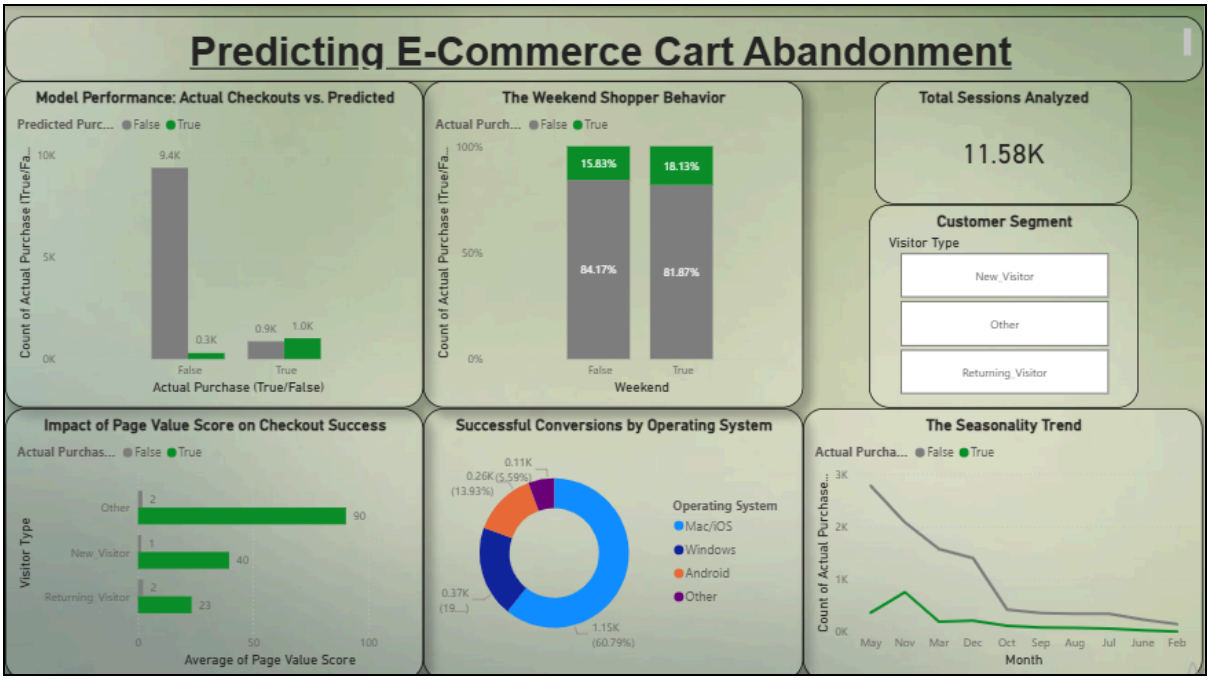
'Partition'	1_Training		2_Testing	
Correct	6,802	92.9%	2,947	92.18%
Wrong	520	7.1%	250	7.82%
Total	7,322		3,197	





4. Executive Storytelling & Business Intelligence (Power BI)

While SPSS handles the complex mathematical modeling, Power BI was chosen to translate those machine learning outputs into an interactive, easy-to-understand executive dashboard. This bridges the gap between raw data science and business strategy, allowing stakeholders to easily interact with the data and make quick, data-driven marketing decisions.



Dashboard Components Explained:

- Model Performance: A bar chart comparing actual checkouts vs. predicted outcomes, demonstrating the SPSS model's accuracy to business executives.
- Impact of Page Value Score: A horizontal chart visually proving the SPSS model's primary finding—that higher Page Values strongly correlate with successful checkouts.
- Conversions by OS: A donut chart highlighting that Mac/iOS users drive the highest volume of successful purchases.
- Behavioral Trends: Visuals tracking 'Weekend Shopper Behavior' and 'Seasonality Trends' to help marketing teams optimize their ad campaigns.
- Interactive Slicer: A 'Customer Segment' slicer allowing users to filter the entire dashboard by New vs. Returning visitors.